Iniciación práctica al análisis de datos OMOP

El papel de los datos del mundo real para generar evidencia en Salud Pública:

El ejemplo de la pandemia COVID-19

Elena Roel Herranz
eroel@idiapjgol.org

Conclusiones

Durante la pandemia, los datos del mundo real han sido claves para generar evidencia y guiar los esfuerzos en Salud Pública, gracias a sus características:

Conclusiones

Durante la pandemia, los datos del mundo real han sido claves para generar evidencia y guiar los esfuerzos en Salud Pública, gracias a sus características:

Disponibilidad

Conclusiones

Durante la pandemia, los datos del mundo real han sido claves para generar evidencia y guiar los esfuerzos en Salud Pública, gracias a sus características:

Disponibilidad	Volumen de datos

Conclusiones

Durante la pandemia, los datos del mundo real han sido claves para generar evidencia y guiar los esfuerzos en Salud Pública, gracias a sus características:

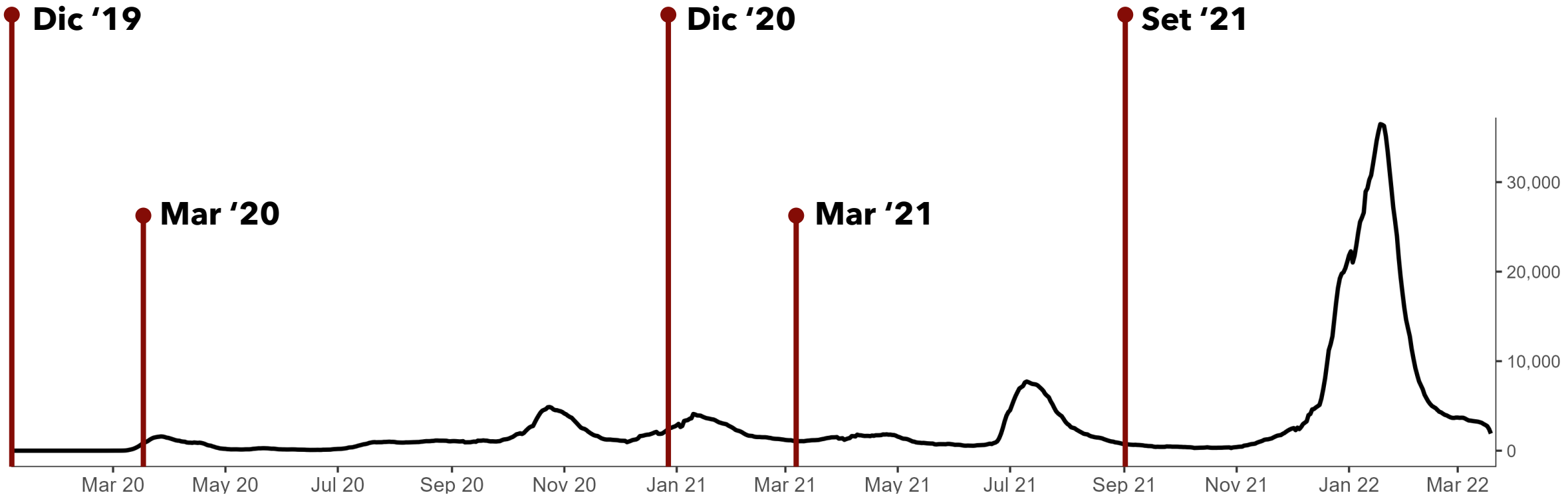
Disponibilidad	Volumen de datos
Representatividad	

Conclusiones

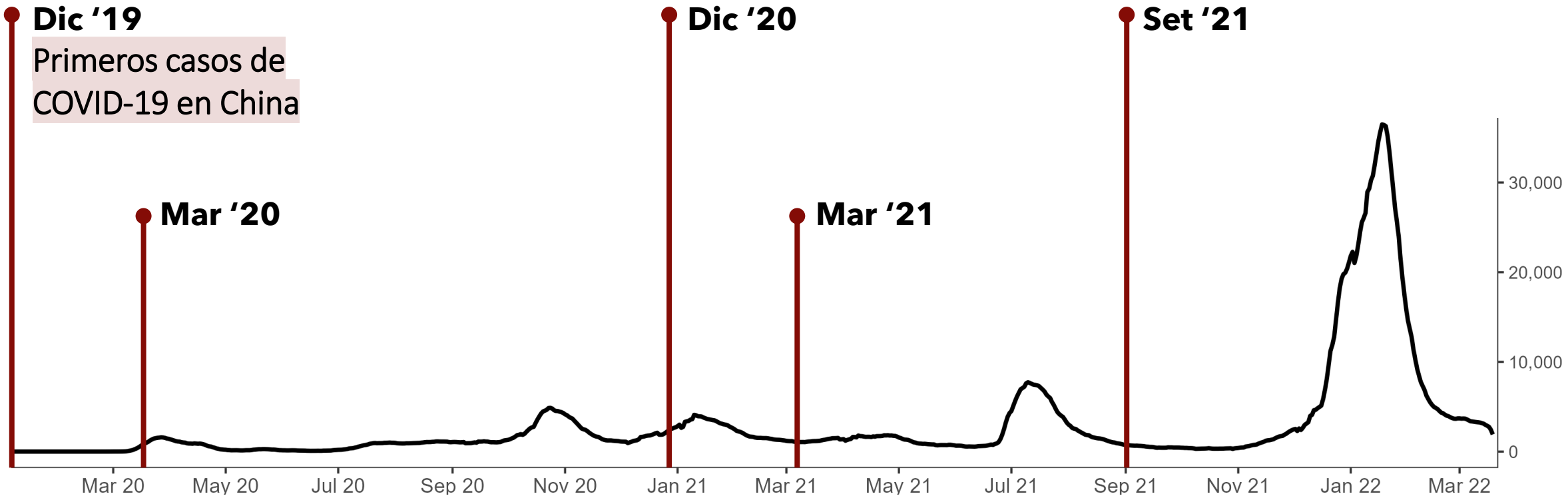
Durante la pandemia, los datos del mundo real han sido claves para generar evidencia y guiar los esfuerzos en Salud Pública, gracias a sus características:

- Disponibilidad
- Volumen de datos
- Representatividad
- Heterogeneidad

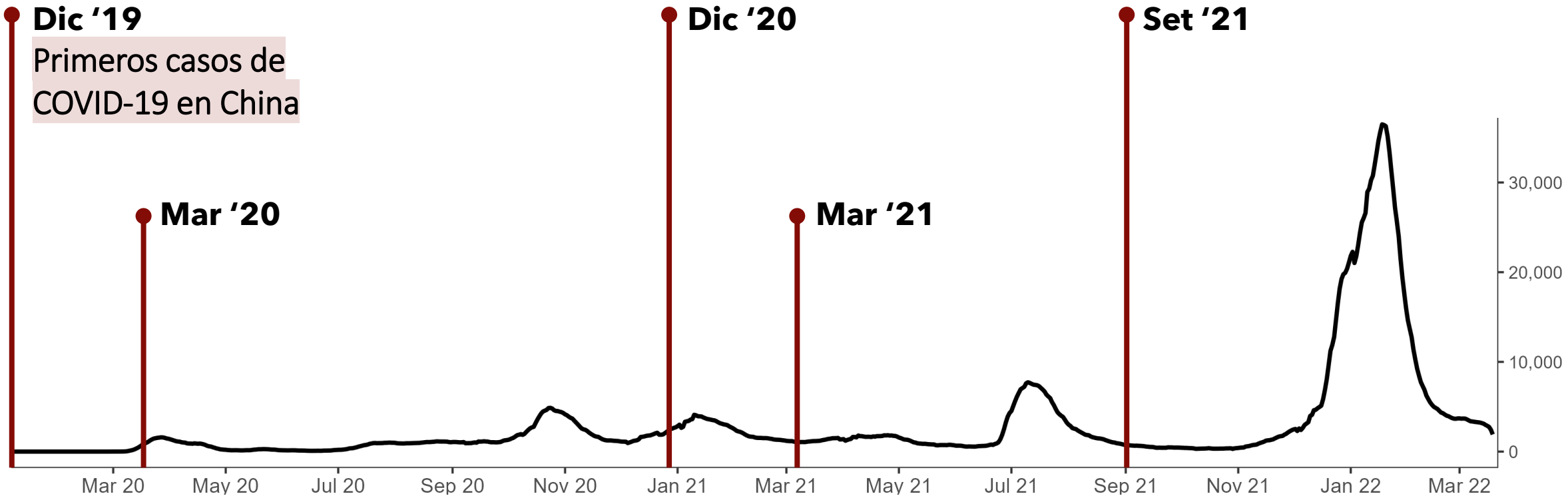
Índice


Dic ‘19
Mar ‘20
Dic ‘20
Mar ‘21
Set ‘21
30,000
20,000
10,000
0
Mar 20
May 20
Jul 20
Sep 20
Nov 20
Jan 21
Mar 21
May 21
Jul 21
Sep 21
Nov 21
Jan 22
Mar 22

Índice


Dic ‘19
Primeros casos de COVID-19 en China
Mar ‘20
Dic ‘20
Mar ‘21
Set ‘21
30,000
20,000
10,000
0
Mar 20
May 20
Jul 20
Sep 20
Nov 20
Jan 21
Mar 21
May 21
Jul 21
Sep 21
Nov 21
Jan 22
Mar 22

1. ¿Qué características tienen las personas con COVID-19?

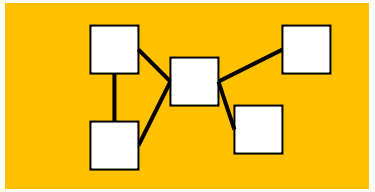


Proyecto CHARYBDIS - OHDSI

Caracterización a gran escala de pacientes con COVID-19:

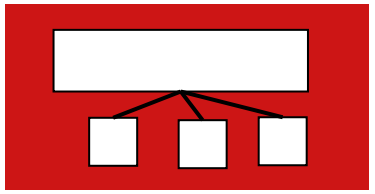
- 25 bases de datos estandarizadas a OMOP CDM de 9 países
- 4.5 M personas con un diagnostico de COVID-19 y 800,000 hospitalizadas
- >22,000 características descritas

Source 1 raw data



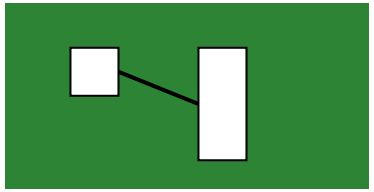
Electronic health records

Source 2 raw data

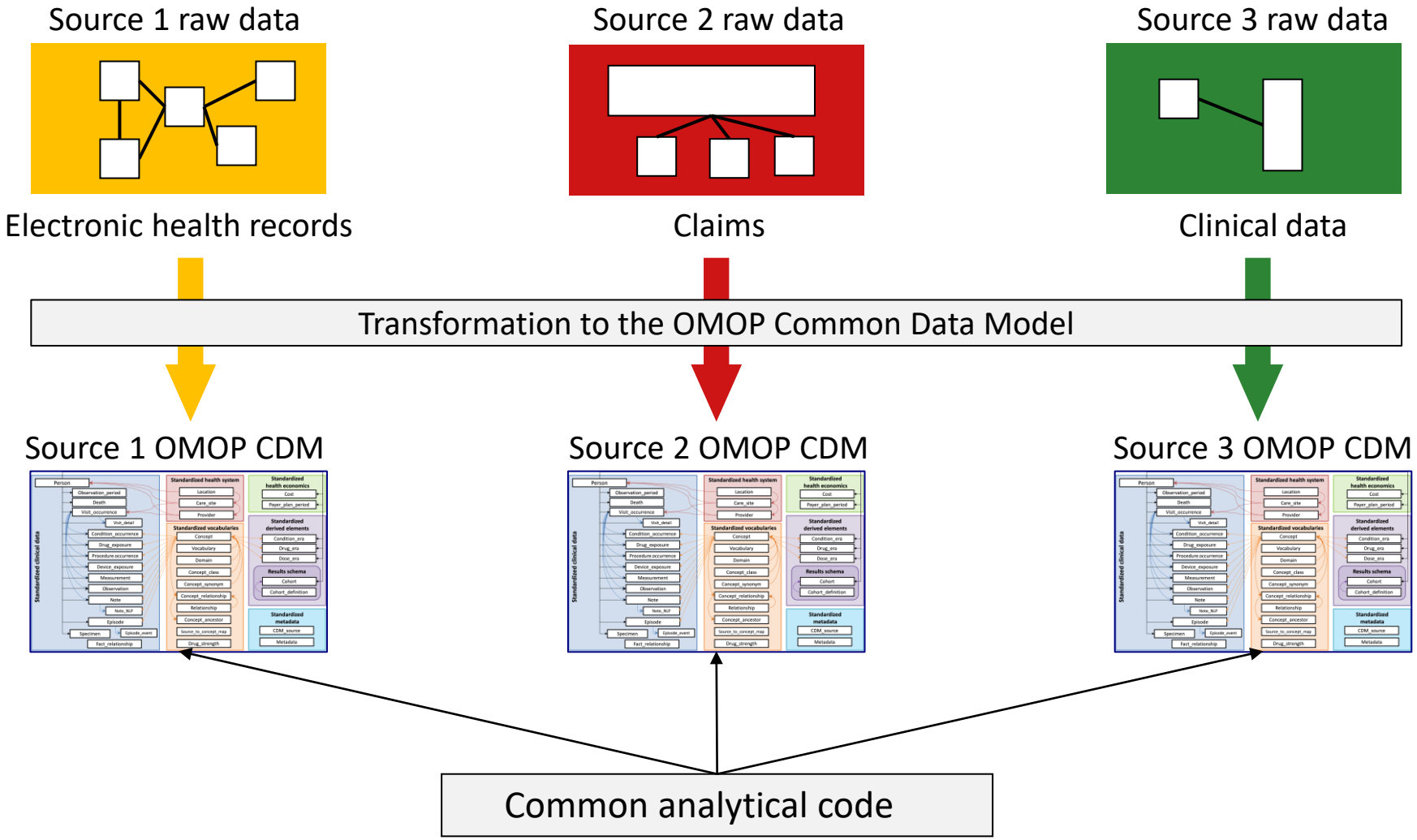


Claims

Source 3 raw data



Clinical data


Source 1 raw data
Source 2 raw data
Source 3 raw data
Electronic health records
Claims
Clinical data
Transformation to the OMOP Common Data Model
Source 1 OMOP CDM
Source 2 OMOP CDM
Source 3 OMOP CDM
Common analytical code

Publicaciones

Oct'20

ARTICLE

https://doi.org/10.1038/s41467-020-18849-z OPEN

Deep phenotyping of 34,128 adult patients hospitalised with COVID-19 in an international network study

May'21

RESEARCH

Use of repurposed and adjuvant drugs in hospital patients with covid-19: multinational network cohort study

Jul'21

ARTICLE OPEN

Epidemiology and Population Health

Characteristics and outcomes of 627 044 COVID-19 patients living with and without obesity in the United States, Spain, and the United Kingdom

Sep'21

Thirty-Day Outcomes of Children and Adolescents With COVID-19: An International Experience

Oct'21

COVID-19 in patients with autoimmune diseases: characteristics and outcomes in a multinational network of cohorts across three countries

Oct'21

CANCER EPIDEMIOLOGY, BIOMARKERS & PREVENTION | RESEARCH ARTICLE

Characteristics and Outcomes of Over 300,000 Patients with COVID-19 and History of Cancer in the United States and Spain

Dec'21

BMJ Open Characteristics and outcomes of patients with COVID-19 with and without prevalent hypertension: a multinational cohort study

Mar'22

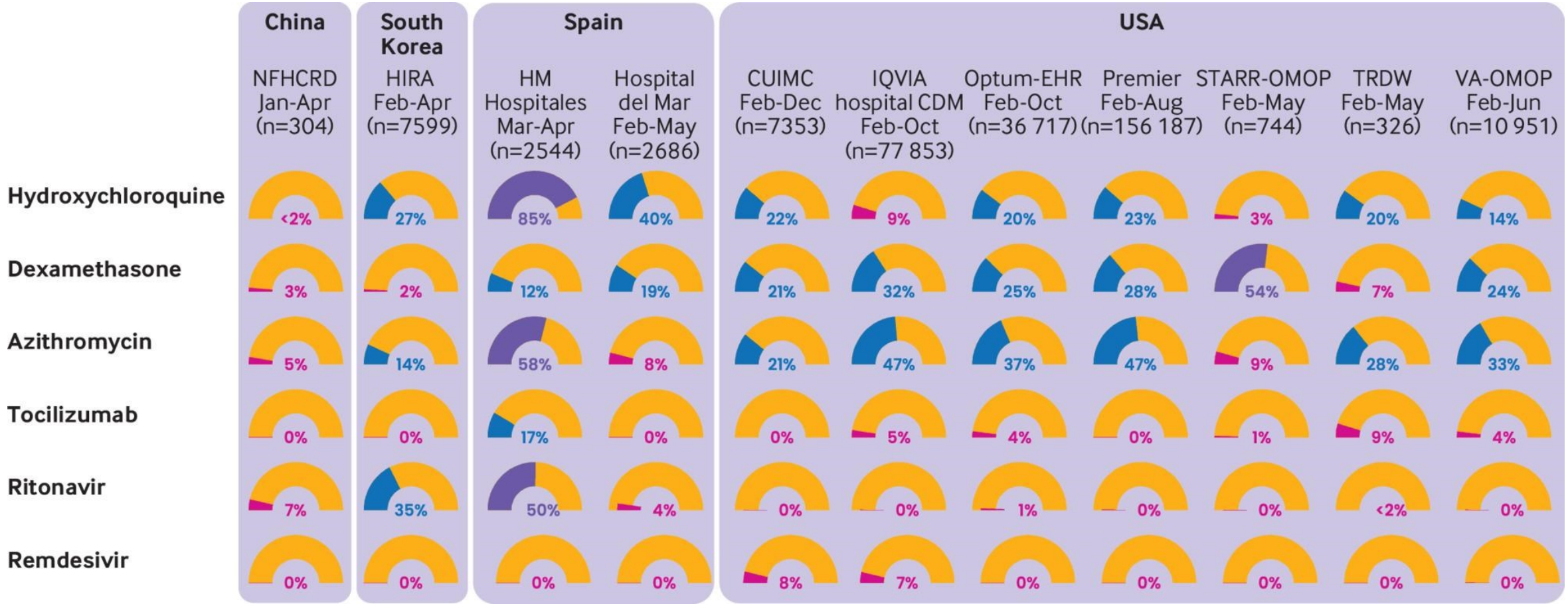
ORIGINAL RESEARCH

Unraveling COVID-19: A Large-Scale Characterization of 4.5 Million COVID-19 Cases Using CHARYBDIS

Jan'23

Characteristics and outcomes of COVID-19 patients with and without asthma from the United States, South Korea, and Europe

Porcentaje de uso de medicamentos entre 0-30 días después del ingreso en pacientes con COVID-19



Otro ejemplo: AESIs y COVID-19

EClinicalMedicine. 2023 Apr:58:101932. doi: 10.1016/j.eclinm.2023.101932. Epub 2023 Apr 4.

Contextualising adverse events of special interest to characterise the baseline incidence rates in 24 million patients with COVID-19 across 26 databases: a multinational retrospective cohort study

We included 23,840,986 patients with COVID-19 from 26 databases representing a diverse set of care settings from North America, Europe, and Asia (11 countries)

Otro ejemplo: AESIs y COVID-19

Source	SIR (95% CI)
Pulmonary Embolism	11.77 (10.08 – 13.73)
Disseminated Intravascular Coagulation	8.79 (7.13 – 10.85)
Myocarditis Pericarditis	7.70 (6.61 – 8.98)
Guillain Barre Syndrome	6.79 (5.03 – 9.17)
Encephalomyelitis	5.94 (4.59 – 7.70)
Thrombosis with Thrombocytopenia (TWT)	5.60 (4.74 – 6.62)
Deep Vein Thrombosis (DVT)	5.56 (4.57 – 6.76)
Acute Myocardial Infarction	4.70 (4.07 – 5.43)
Non–hemorrhagic Stroke	4.59 (3.95 – 5.34)
Hemorrhagic Stroke	3.86 (3.26 – 4.57)
Immune Thrombocytopenia (ITP)	3.37 (2.96 – 3.82)
Transverse Myelitis	2.53 (2.01 – 3.17)
Anaphylaxis	1.93 (1.56 – 2.40)
Bells Palsy	1.84 (1.60 – 2.12)
Appendicitis	1.72 (1.47 – 2.02)
Narcolepsy	1.32 (1.05 – 1.66)

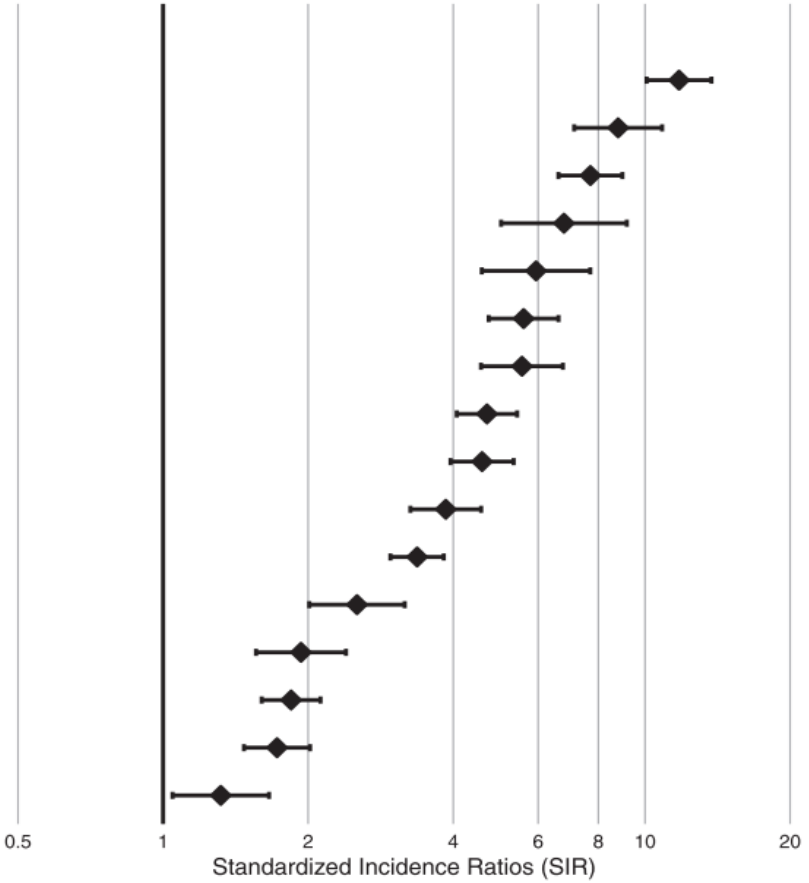
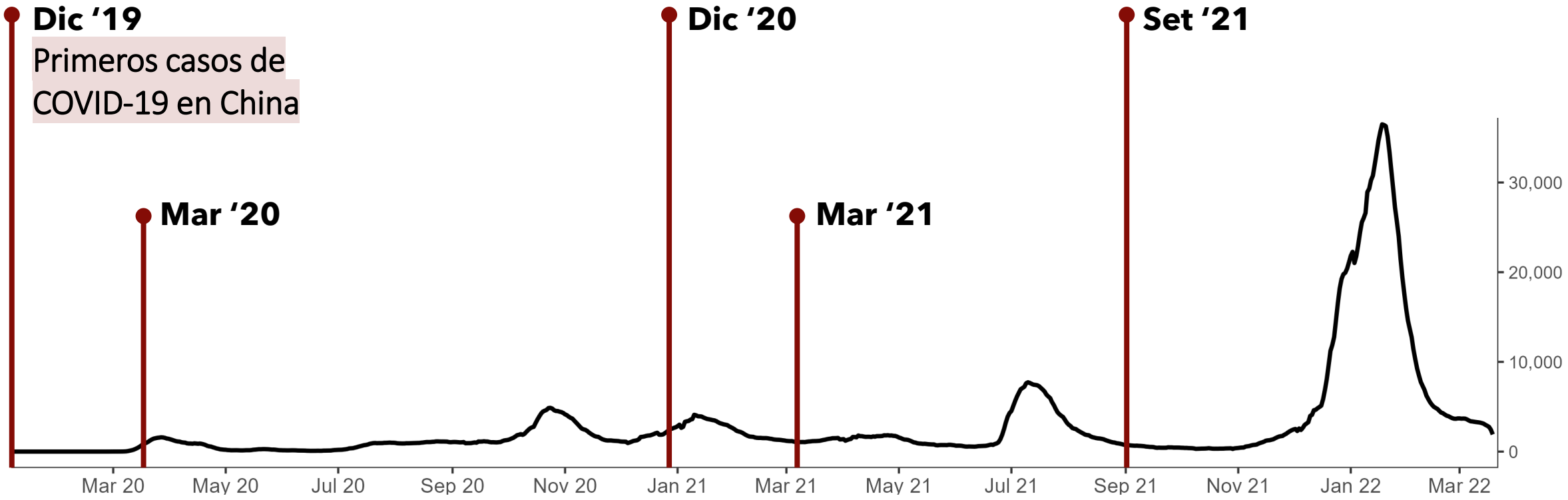


***Fig. 3:* Standardised incidence ratios forest plot with meta analysis comparing the 'patients with COVID-19' to the 'pre-pandemic background population'.** CI = confidence interval.

2. ¿Qué personas tienen más riesgo de COVID-19 y sus complicaciones?



Riesgo de COVID-19 y complicaciones en personas con cáncer

Cancer and the risk of coronavirus disease 2019 diagnosis, hospitalisation and death: A population-based multistate cohort study including 4 618 377 adults in Catalonia, Spain

Elena Roel, Andrea Pistillo, Martina Recalde, Sergio Fernández-Bertolín, María Aragón, Isabelle Soerjomataram, Mazda Jenab, Diana Puente, Daniel Prieto-Alhambra, Edward Burn, Talita Duarte-Salles ... See fewer authors

First published: 16 October 2021 | https://doi.org/10.1002/ijc.33846 | Citations: 15

We investigated the natural history of COVID-19 in cancer patients during the first wave of the pandemic in a large, representative cohort with a heterogenous cancer population.

Fuente de datos: SIDIAP

- Base de datos de historias clínicas electrónicas de AP representativa que incluye al 75% de la población residente en Cataluña (6.1 M people) desde el 2006
- Vinculación a datos de mortalidad, altas hospitalarias(*CMBD*), tests COVID-19 y estandarizados a OMOP CDM
- Participantes: 4,6 M

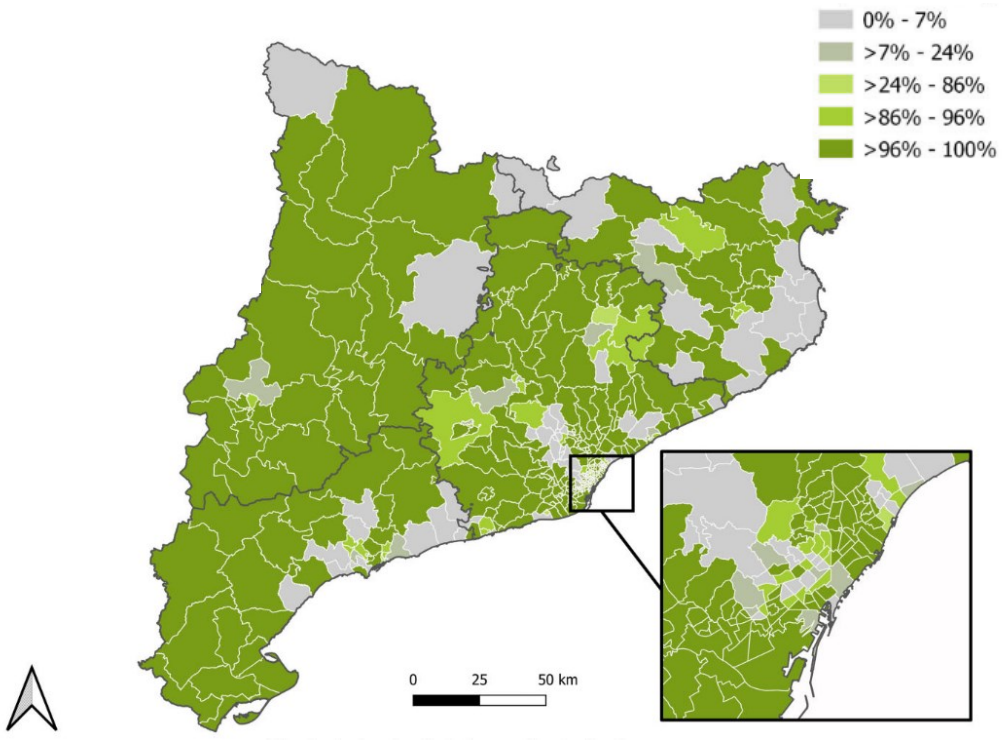


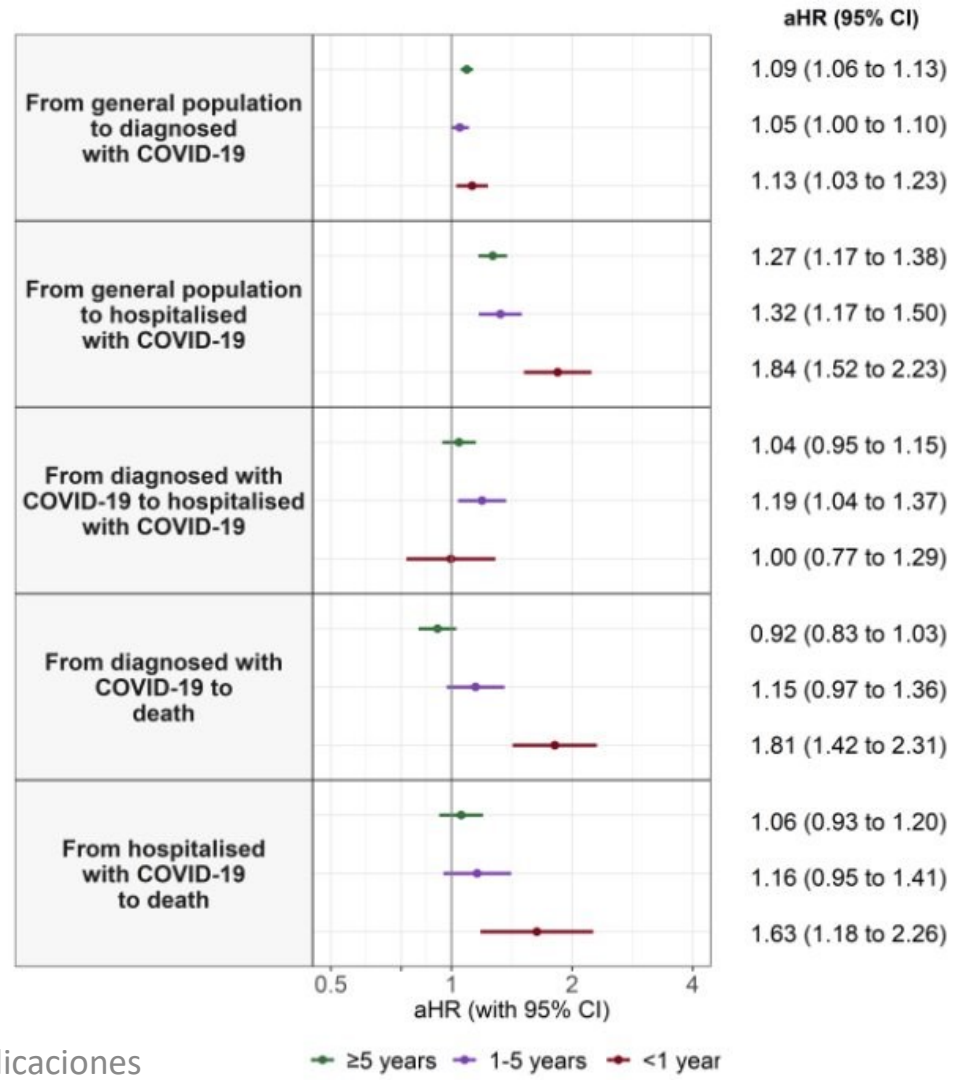
Fig. Cobertura del SIDIAP, 30 de Junio 2021

Source: Recalde et al (2022)

Riesgo de COVID-19 y complicaciones en personas con cáncer

El cáncer se asocia con un mayor riesgo de diagnóstico de COVID-19, hospitalización y muerte post-hospitalización, especialmente en pacientes con un diagnóstico reciente y con cáncer hematológico

HR de eventos Covid-19 en personas con vs sin cáncer



Otro ejemplo: mortalidad y COVID-19 en el Reino Unido

Article | Published: 08 July 2020

Factors associated with COVID-19-related death using OpenSAFELY

Elizabeth J. Williamson, Alex J. Walker, Krishnan Bhaskaran, Seb Bacon, Chris Bates, Caroline E. Morton, Helen J. Curtis, Amir Mehrkar, David Evans, Peter Inglesby, Jonathan Cockburn, Helen I. McDonald, Brian MacKenna, Laurie Tomlinson, Ian J. Douglas, Christopher T. Rentsch, Rohini Mathur, Angel Y. S. Wong, Richard Grieve, David Harrison, Harriet Forbes, Anna Schultze, Richard Croker, John Parry, ... Ben Goldacre

+ Show authors

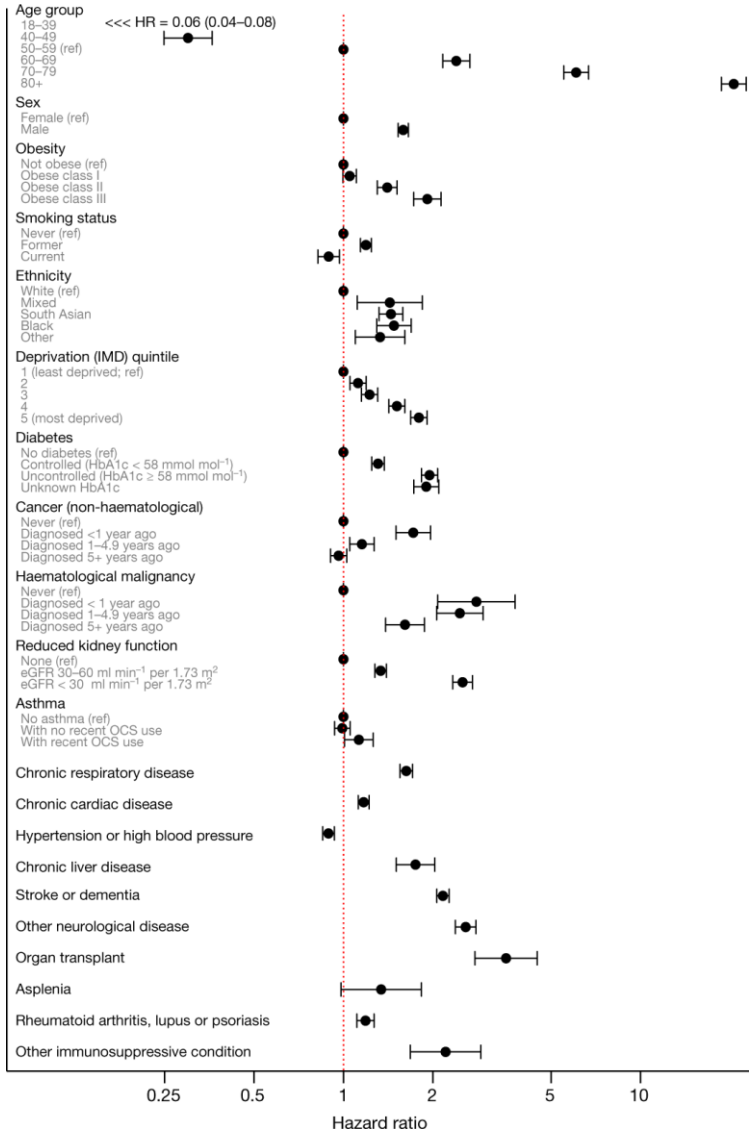
Nature **584**, 430–436 (2020) | Cite this article

Primary care records of 17,278,392 adults were pseudonymously linked to 10,926 COVID-19-related deaths

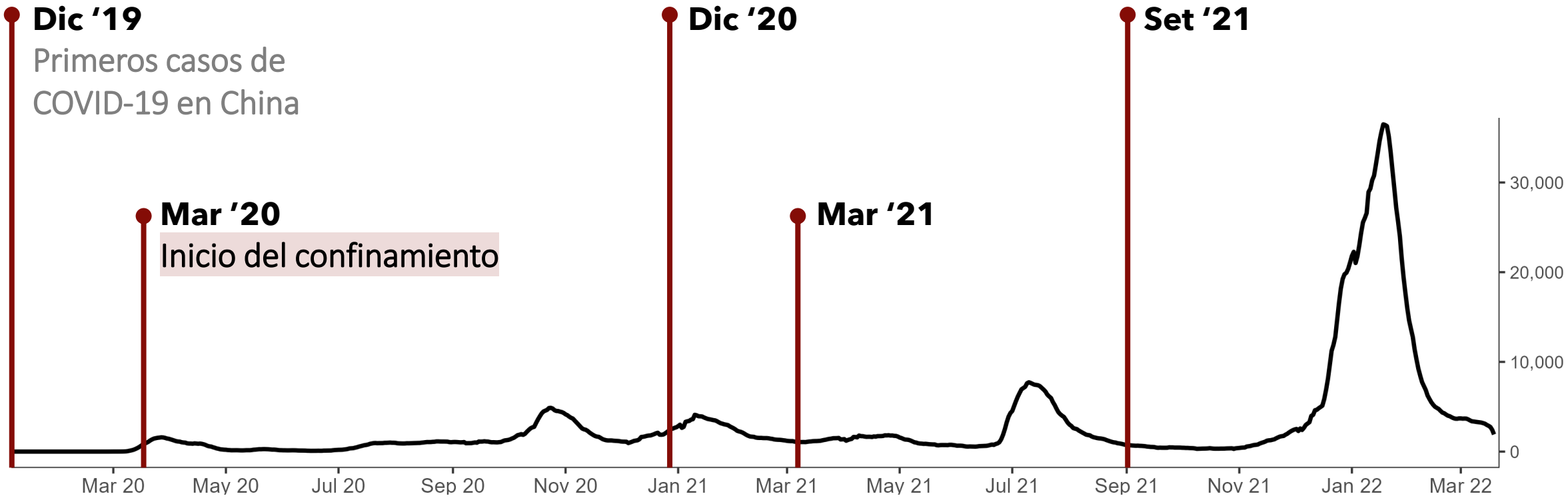
Otro ejemplo: mortalidad y COVID-19 en el Reino Unido

Mortalidad por COVID-19 asociada con sexo masculino, edad, peor nivel socioeconómico, diabetes, asma, …

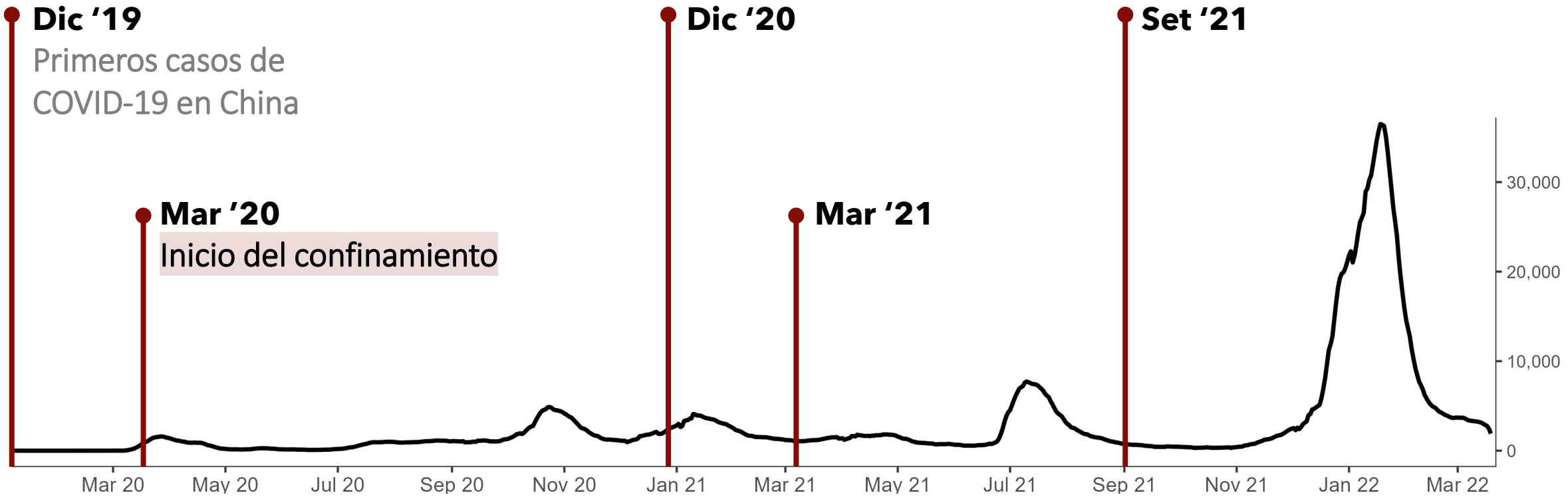
Estimated hazard ratios for each patient characteristic from a multivariable Cox model.



Índice


Dic ‘19
Primeros casos de COVID-19 en China
Mar ‘20
Inicio del confinamiento
Dic ‘20
Mar ‘21
Set ‘21
30,000
20,000
10,000
0
Mar 20
May 20
Jul 20
Sep 20
Nov 20
Jan 21
Mar 21
May 21
Jul 21
Sep 21
Nov 21
Jan 22
Mar 22

3. ¿Qué consecuencias indirectas tuvo el confinamiento/la pandemia?



Impacto en trastornos alimentarios en Cataluña

Impact of the COVID-19 pandemic on eating disorders diagnoses among adolescents and young adults in Catalonia: A population-based cohort study

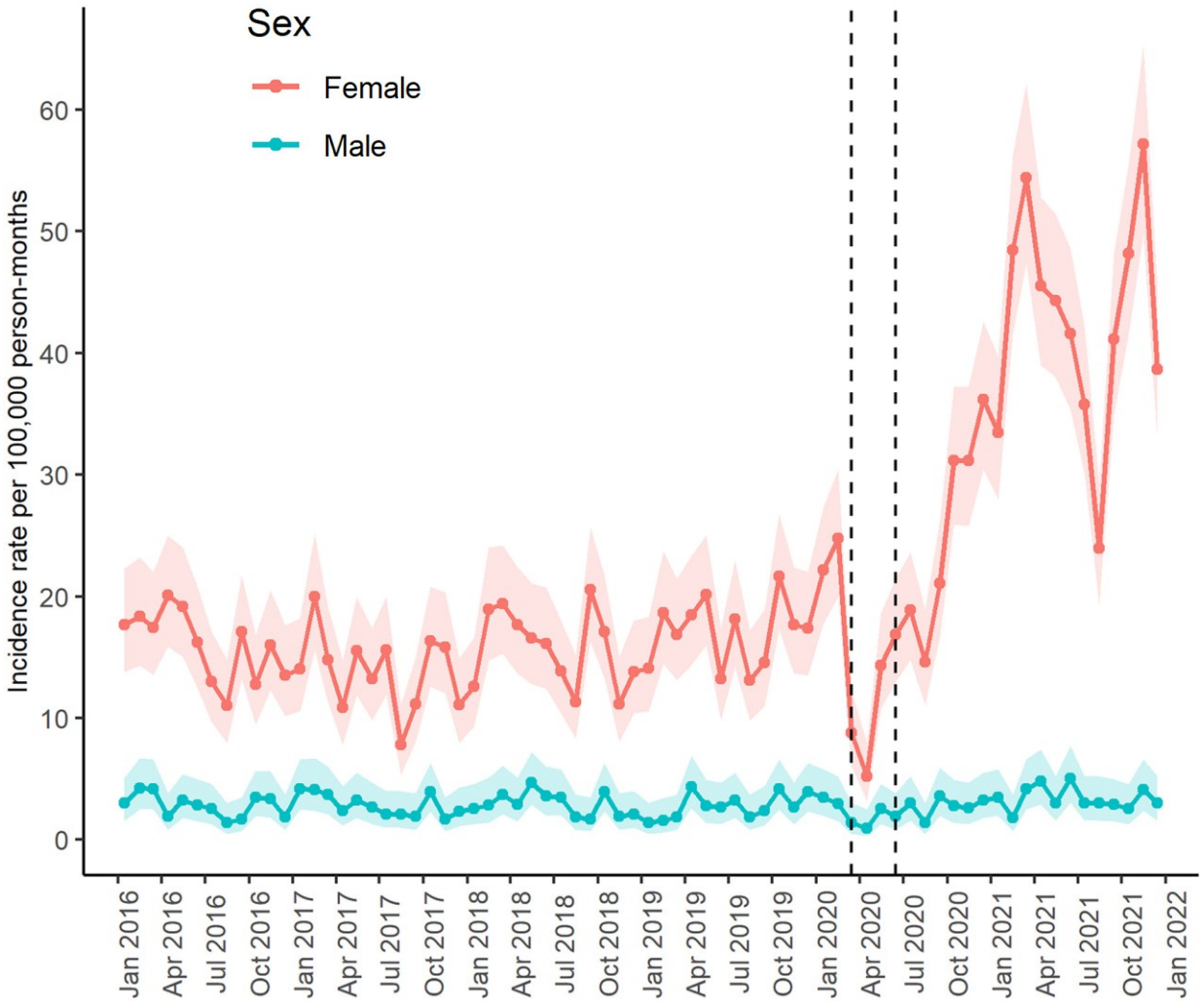
Berta Raventós MSc, Alicia Abellan PhD, Andrea Pistillo MSc, Carlen Reyes PhD, Edward Burn PhD, Talita Duarte-Salles PhD

First published: 09 November 2022 | https://doi.org/10.1002/eat.23848

Population-based cohort study using primary care records of 1,2M people aged 10-24 years between January, 2016 and December, 2021 in Catalonia, Spain.

Impacto en trastornos alimentarios en Cataluña

Aumento del 110% en la incidencia de diagnósticos de trastornos alimentarios en chicas de 10-24 años



Otro ejemplo: impacto en diagnósticos de cáncer

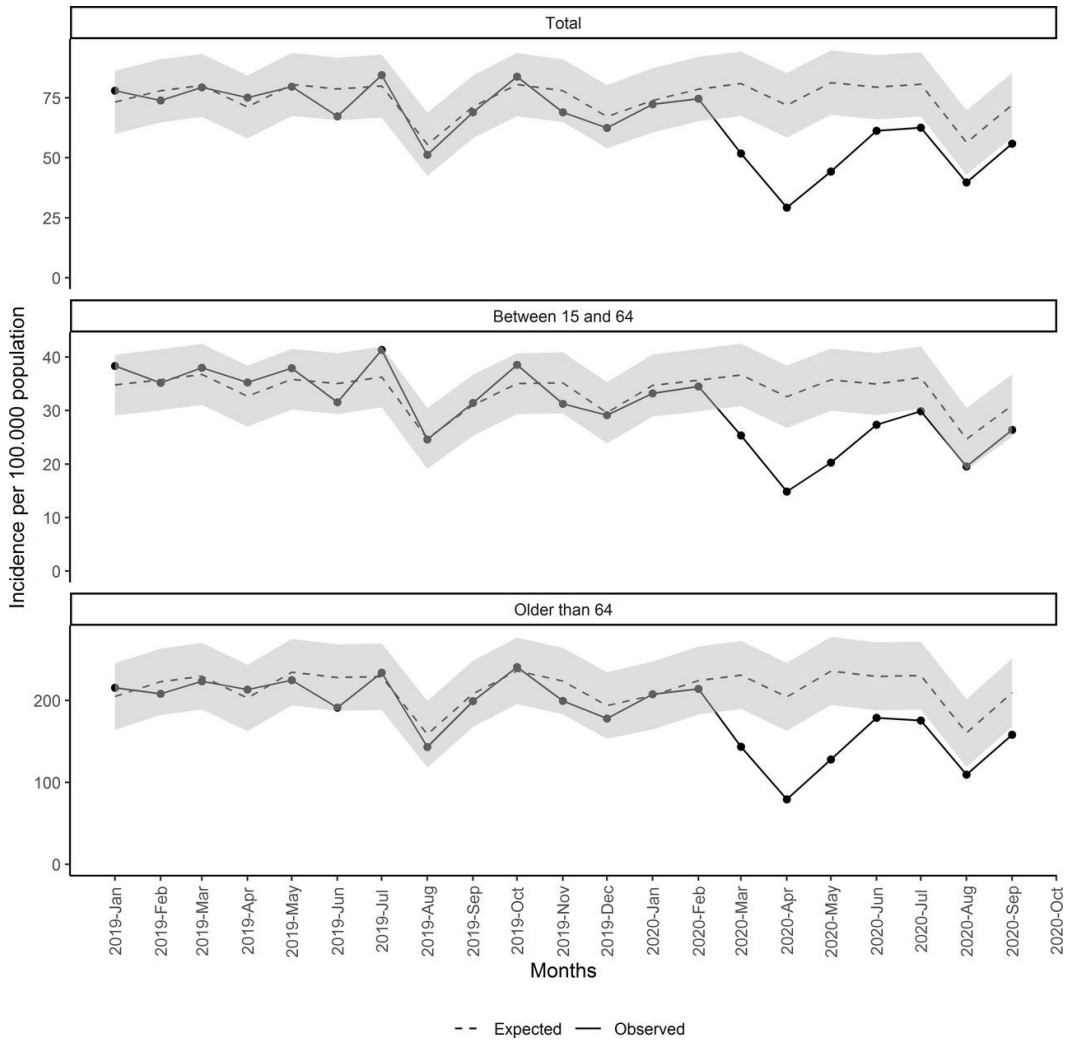
Impact of the COVID-19 pandemic and related control measures on cancer diagnosis in Catalonia: a time-series analysis of primary care electronic health records covering about five million people

Ermengol Coma [1], Carolina Guiriguet [1, 2], Nuria Mora [1, 3], Mercè Marzo-Castillejo [3, 4], Mencia Benítez [1, 2], Leonardo Méndez-Boo [1], Francesc Fina [1], Mireia Fàbregas [1], Albert Mercadé [1], Manuel Medina [1]

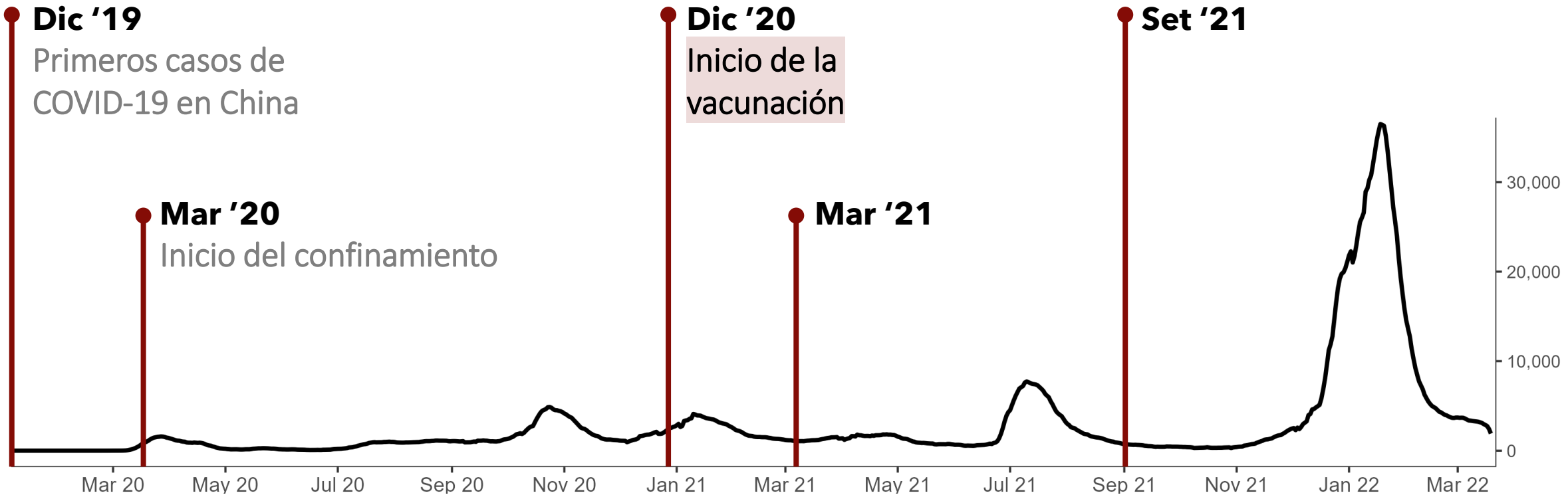
Time-series study of malignant neoplasms and diagnostic procedures, using data from the primary care electronic health records from January 2014 to September 2020.

Otro ejemplo: impacto en diagnósticos de cáncer

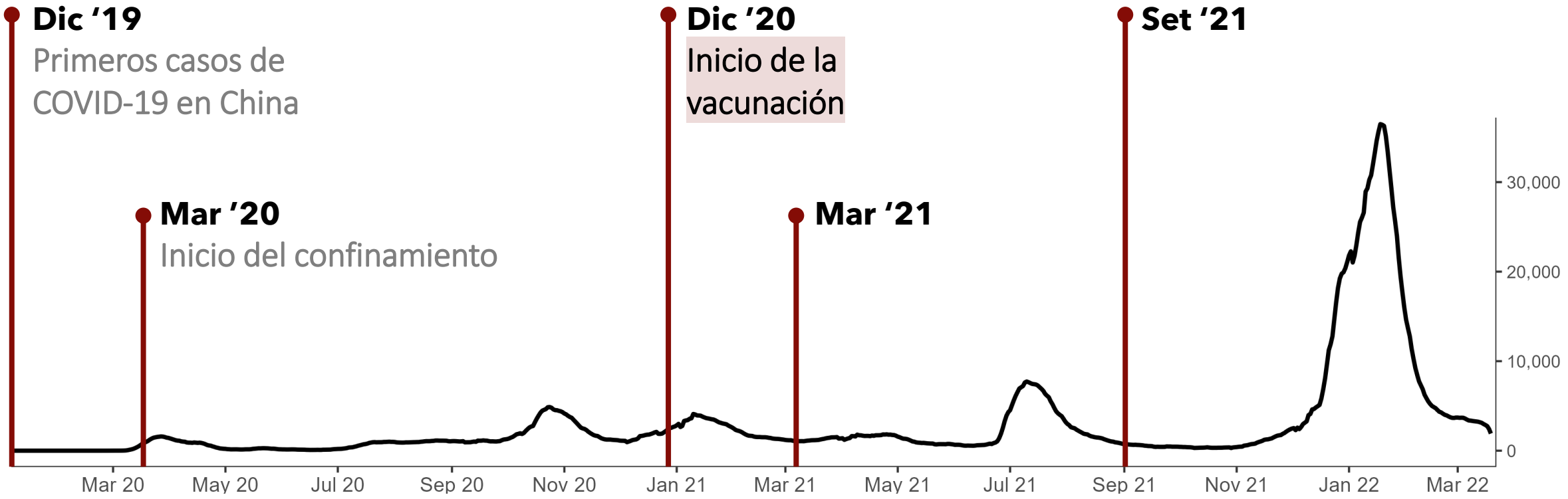
Reducción del 34%
en la incidencia de
diagnósticos de cáncer



Índice


Dic ‘19
Primeros casos de COVID-19 en China
Mar ‘20
Inicio del confinamiento
Dic ‘20
Inicio de la vacunación
Mar ‘21
Set ‘21
30,000
20,000
10,000
0
Mar 20
May 20
Jul 20
Sep 20
Nov 20
Jan 21
Mar 21
May 21
Jul 21
Sep 21
Nov 21
Jan 22
Mar 22

4. Cual es la efectividad de las vacunas COVID-19?



Efectividad de la vacuna contra el COVID-19 en Israel

ORIGINAL ARTICLE

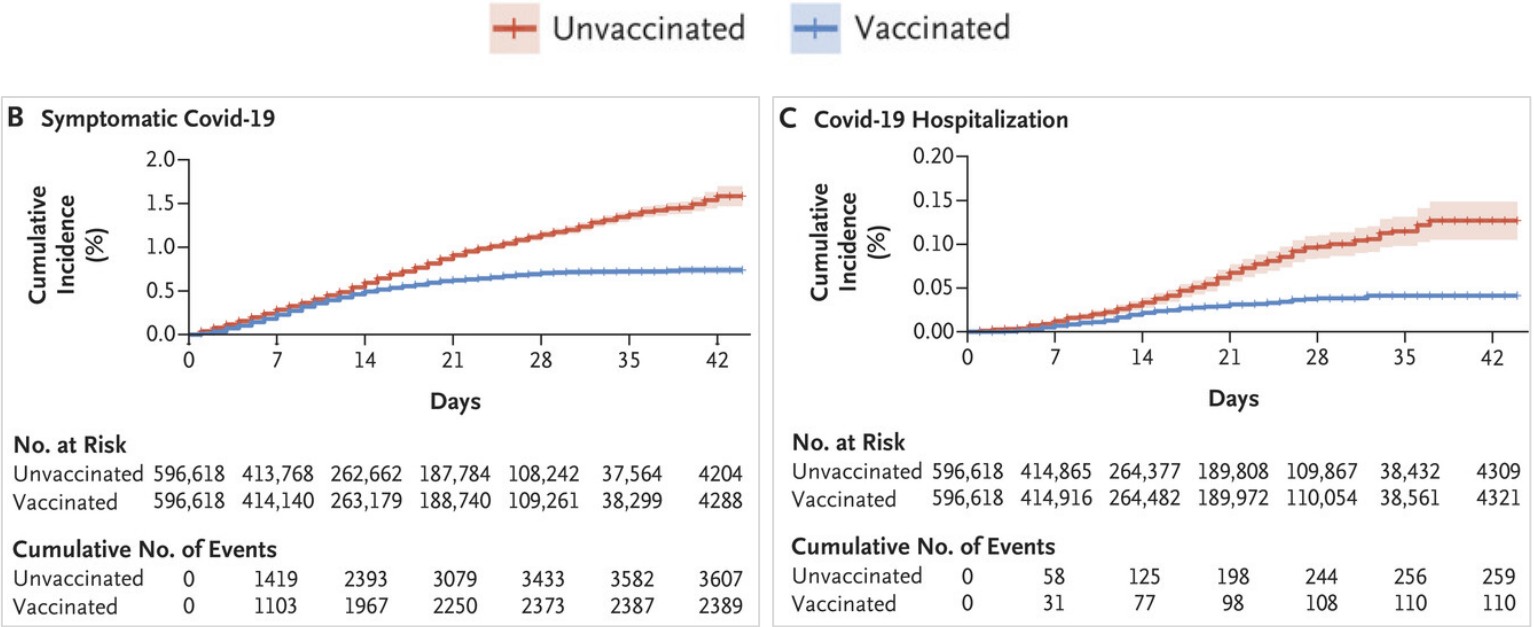
BNT162b2 mRNA Covid-19 Vaccine in a Nationwide Mass Vaccination Setting

Authors: Noa Dagan, M.D., Noam Barda, M.D., Eldad Kepten, Ph.D., Oren Miron, M.A., Shay Perchik, M.A., Mark A. Katz, M.D., Miguel A. Hernán, M.D., Marc Lipsitch, D.Phil., Ben Reis, Ph.D., and Ran D. Balicer, M.D. Author Info & Affiliations

Published February 24, 2021 | N Engl J Med 2021;384:1412-1423 | DOI: 10.1056/NEJMoa2101765

- Fuente de datos: principal proveedor sanitarios de Israel (53% de la población). Incluye datos de primaria, especialistas, hospitalizaciones, pruebas de laboratorio, etc.
- 596,618 parejas de vacunados vs no vacunados

Efectividad de la vacuna contra el COVID-19 en Israel



Efectividad ≥7días tras 2º dosis:

- Infección 92% [88 to 95]
- Infección sintomática 94% [87 to 98]
- Hospitalización 87% [55 to 100]
- Enfermedad severa 92% [75 to 100]

Otros ejemplos

Research

Associations of BNT162b2 vaccination with SARS-CoV-2 infection and hospital admission and death with covid-19 in nursing homes and healthcare workers in Catalonia: prospective cohort study

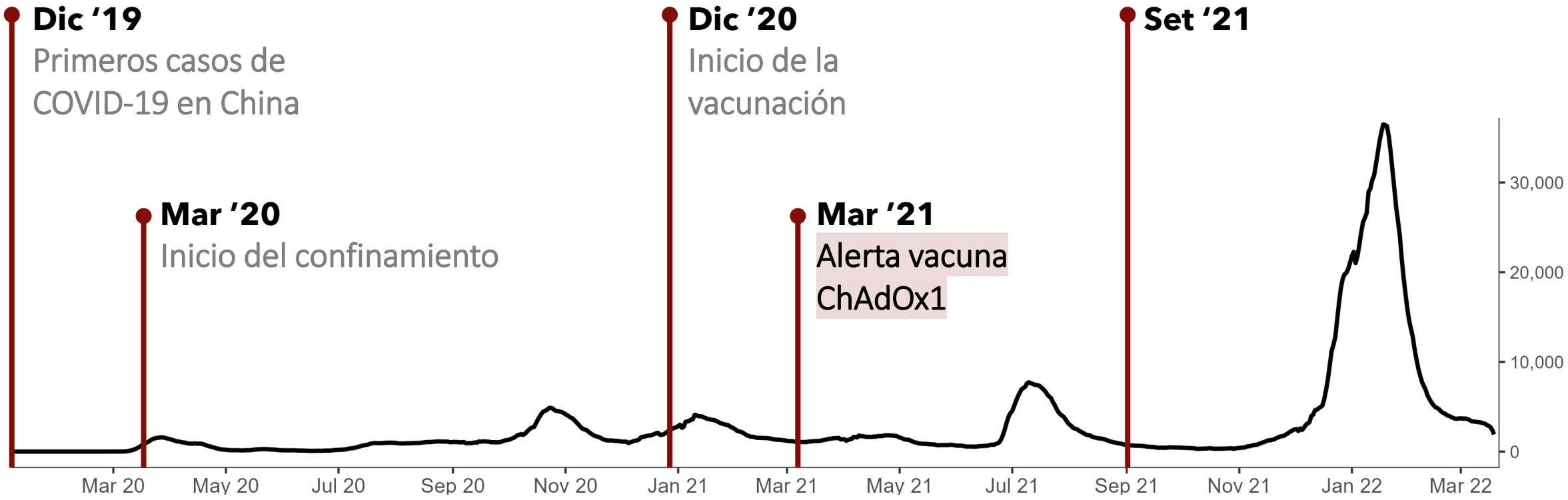
BMJ 2021 ; 374 doi: https://doi.org/10.1136/bmj.n1868 (Published 18 August 2021)
Cite this as: *BMJ* 2021;374:n1868

Effectiveness of COVID-19 vaccines against severe COVID-19 among patients with cancer in Catalonia, Spain

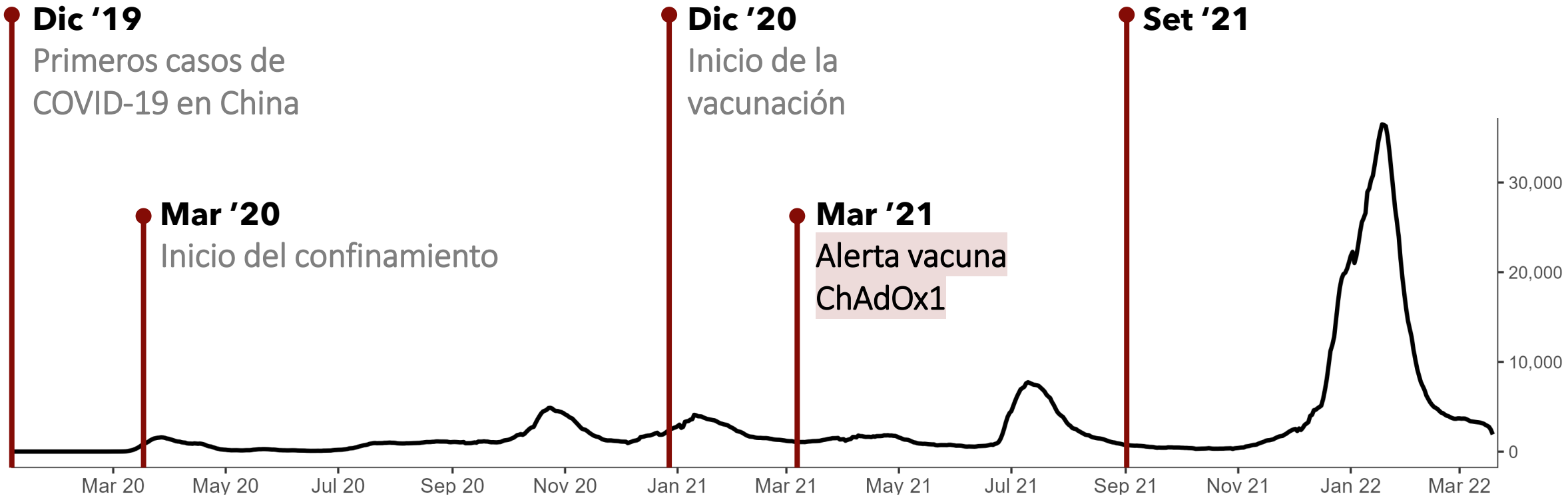
Received: 2 November 2023
Accepted: 28 May 2024

Felippe Lazar Neto[1,2], Núria Mercadé-Besora[3], Berta Raventós[3,4], Laura Pérez-Crespo[3], Gilberto Castro Junior[2], Otavio T. Ranzani[1,5,7] & Talita Duarte-Salles[3,6,7]

Índice


Dic ‘19
Primeros casos de COVID-19 en China
Mar ‘20
Inicio del confinamiento
Dic ‘20
Inicio de la vacunación
Mar ‘21
Alerta vacuna ChAdOx1
Set ‘21
30,000
20,000
10,000
0
Mar 20
May 20
Jul 20
Sep 20
Nov 20
Jan 21
Mar 21
May 21
Jul 21
Sep 21
Nov 21
Jan 22
Mar 22

5. ¿Son seguras las vacunas COVID-19?



Riesgo de trombosis tras infección/vacunación

Thrombosis and thrombocytopenia after vaccination against and infection with SARS-CoV-2 in Catalonia, Spain

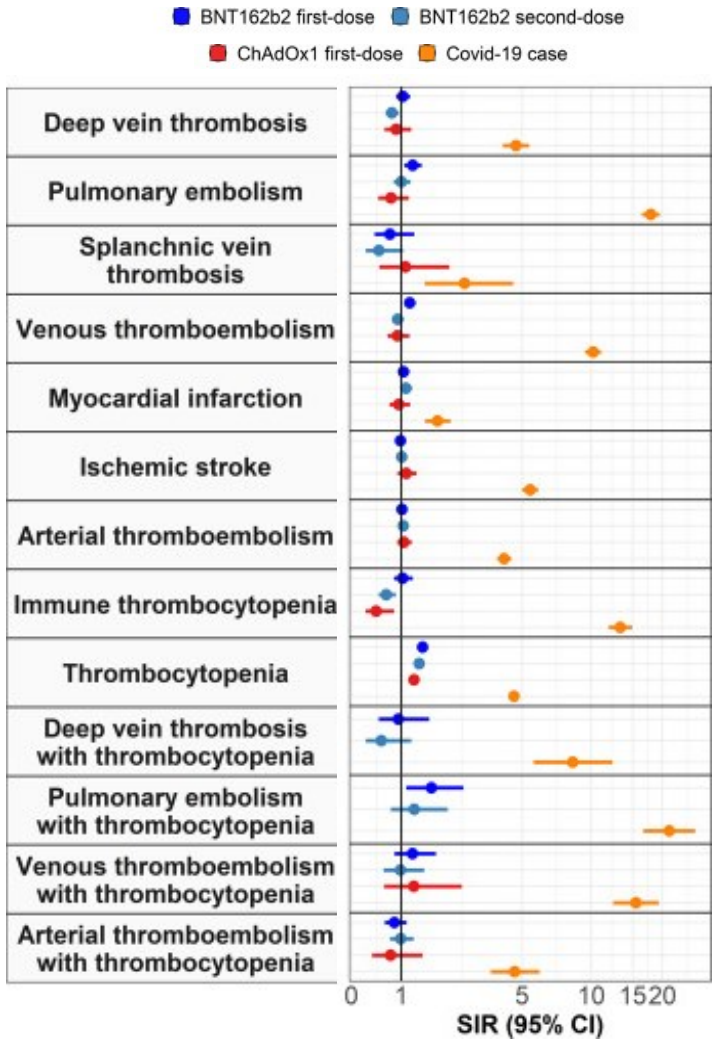
Edward Burn, Elena Roel, Andrea Pistillo, Sergio Fernández-Bertolín, Maria Aragón, Berta Raventós, Carlen Reyes, Katia Verhamme, Peter Rijnbeek, Xintong Li, Victoria Y. Strauss, Daniel Prieto-Alhambra & Talita Duarte-Salles

- *We compare rates of thrombosis and thrombocytopenia following COVID-19 vaccination and infection with the background (expected) rates in the general population.*
- Primary care (SIDIAP) and linked hospital data →
  2M and 590K vaccinated (BN162b2 & ChAdOx1),
  174K COVID-19 cases

Riesgo de trombosis tras infección/vacunación

La incidencia de eventos tromboembólicos venosos, arteriales y de trombocitopenia fue 10, 4 y 5 veces superior a lo esperado en la cohorte COVID-19

SIR de eventos tromboembolicos tras la vacunación/infección COVID-19



Otros ejemplos

Association between covid-19 vaccination, SARS-CoV-2 infection, and risk of immune mediated neurological events: population based cohort and self-controlled case series analysis

BMJ 2022 ; 376 doi: https://doi.org/10.1136/bmj-2021-068373 (Published 16 March 2022)

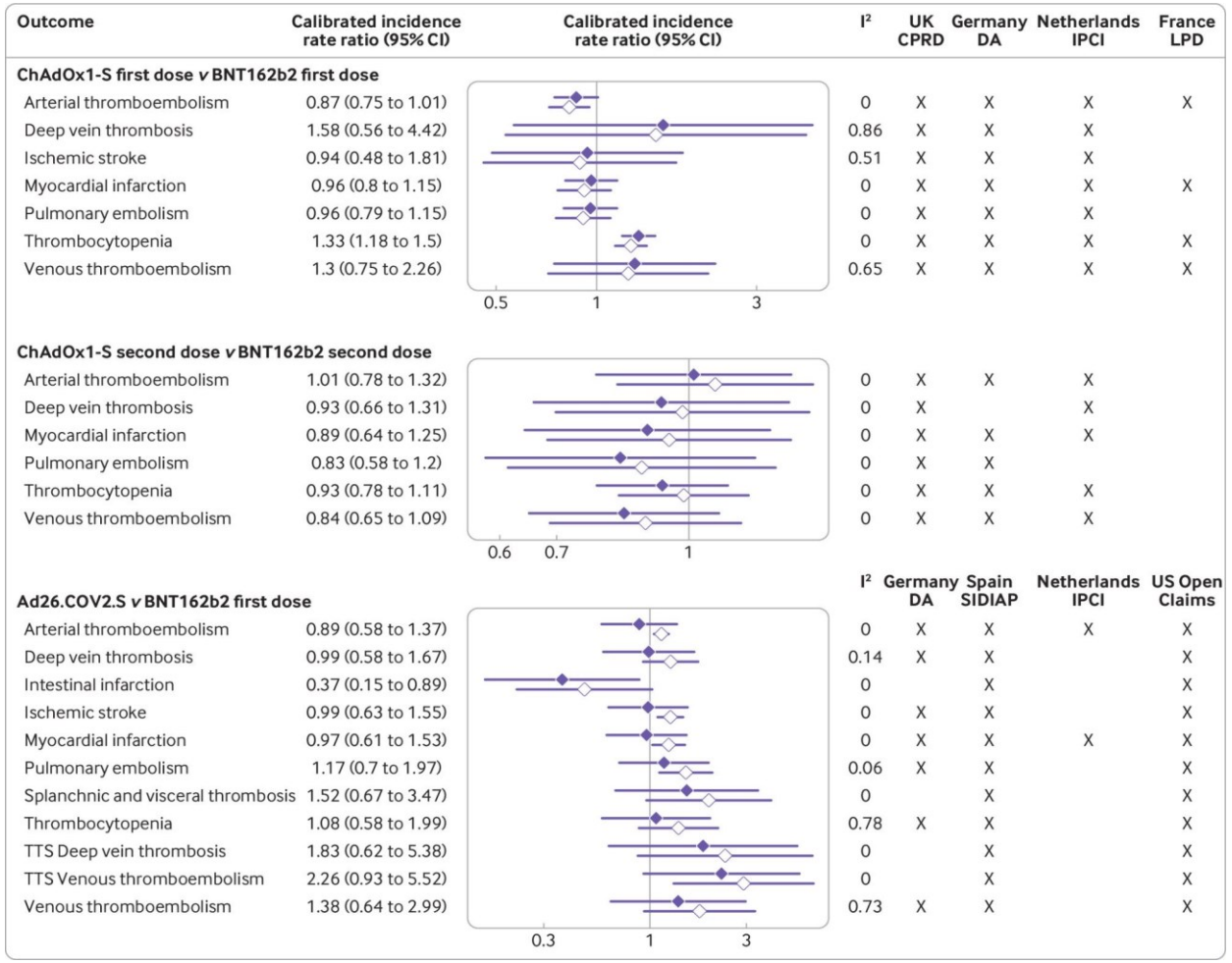
Comparative risk of thrombosis with thrombocytopenia syndrome or thromboembolic events associated with different covid-19 vaccines: international network cohort study from five European countries and the US

BMJ 2022 ; 379 doi: https://doi.org/10.1136/bmj-2022-071594 (Published 26 October 2022)

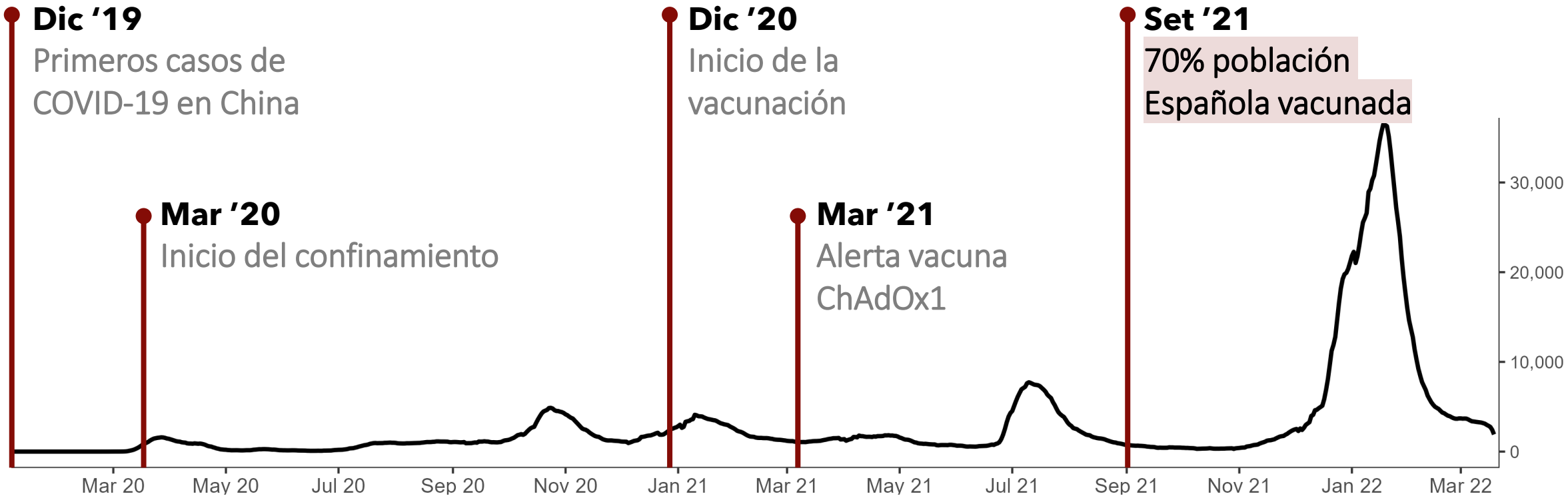
Riesgo de trombosis comparativo entre vacunas

Aumento del 30% en la incidencia de trombocitopenia tras vacunación con ChAdOx1 en comparación con BNT162b2

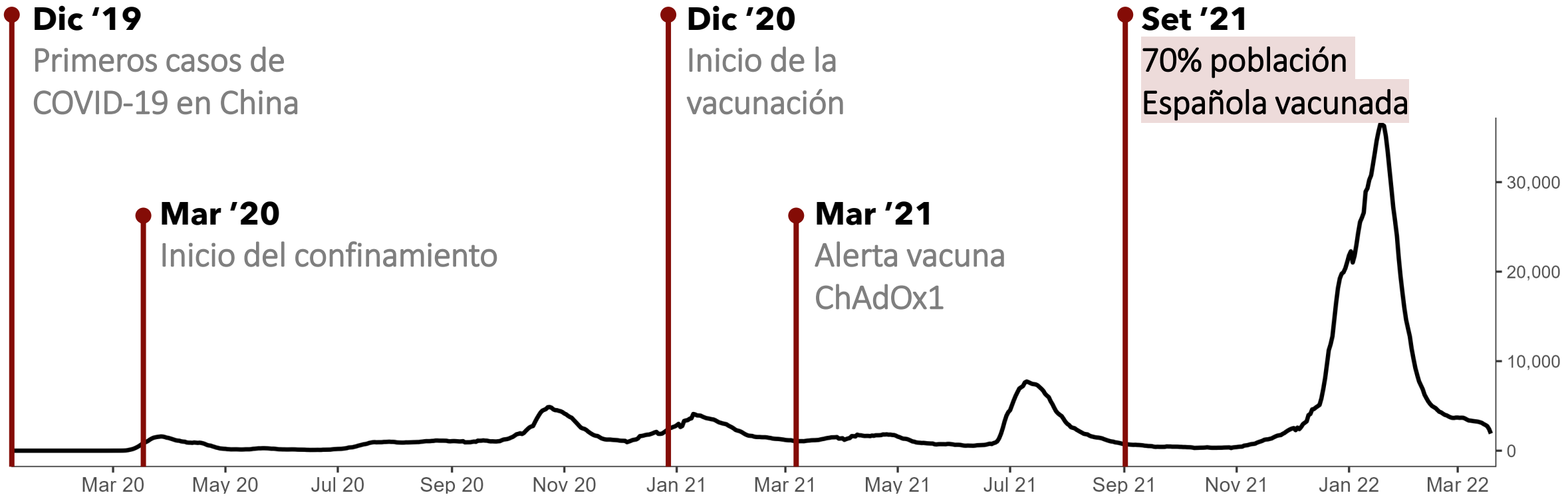
Meta-analytical IRR in the 28 days after vaccination.
Lines with solid diamonds=calibrated estimates;
lines with clear diamonds=uncalibrated estimates



Índice


Dic ’19
Primeros casos de COVID-19 en China
Mar ’20
Inicio del confinamiento
Dic ’20
Inicio de la vacunación
Mar ’21
Alerta vacuna ChAdOx1
Set ’21
70% población Española vacunada
30,000
20,000
10,000
0
Mar 20
May 20
Jul 20
Sep 20
Nov 20
Jan 21
Mar 21
May 21
Jul 21
Sep 21
Nov 21
Jan 22
Mar 22

6. ¿Existen desigualdades en la vacunación COVID-19?



Desigualdades en vacunación COVID-19 en Cataluña

Socioeconomic Inequalities in COVID-19 Vaccination and Infection in Adults, Catalonia, Spain

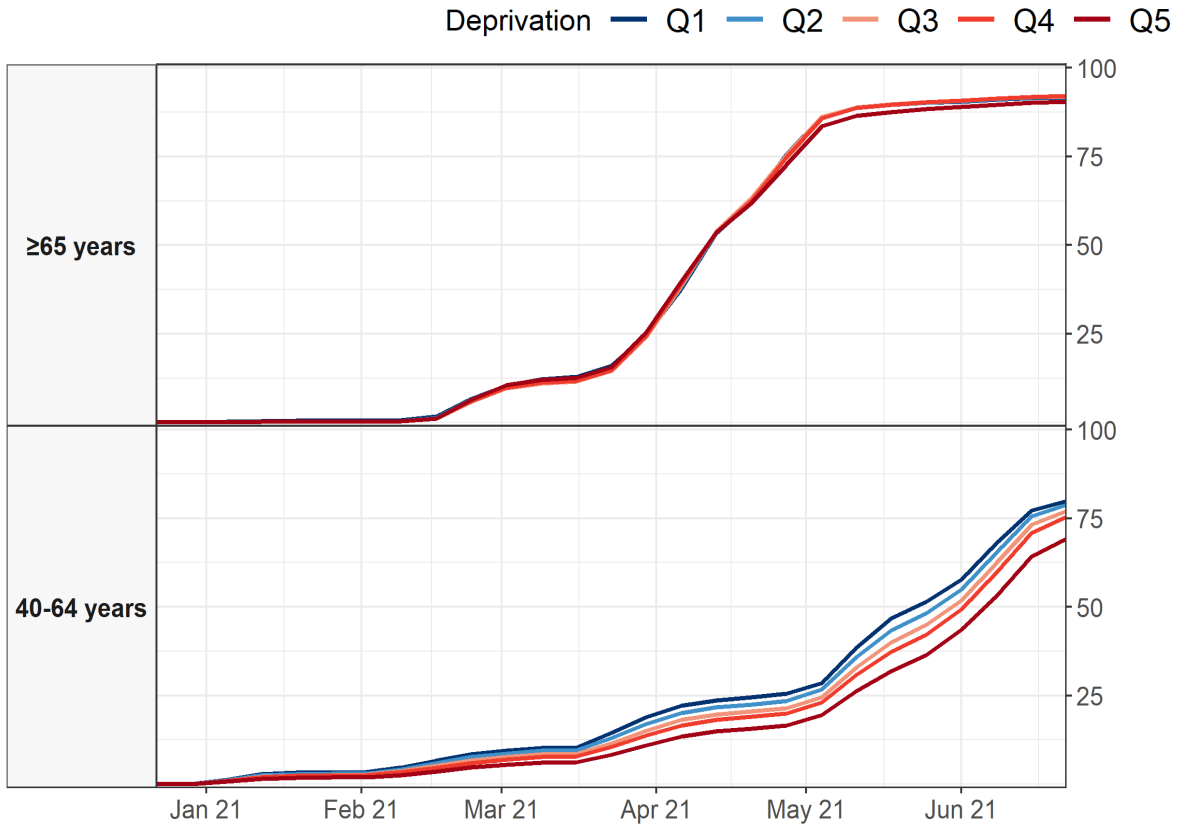
Elena Roel, Berta Raventós, Edward Burn, Andrea Pistillo, Daniel Prieto-Alhambra, Talita Duarte-Salles

- *We conducted a population-based cohort study in Catalonia, Spain, during September 2020–June 2021 that comprised 2,3 M adults >40 years of age*
- SIDIAP data linked to COVID-19 tests, vaccines & hospital records

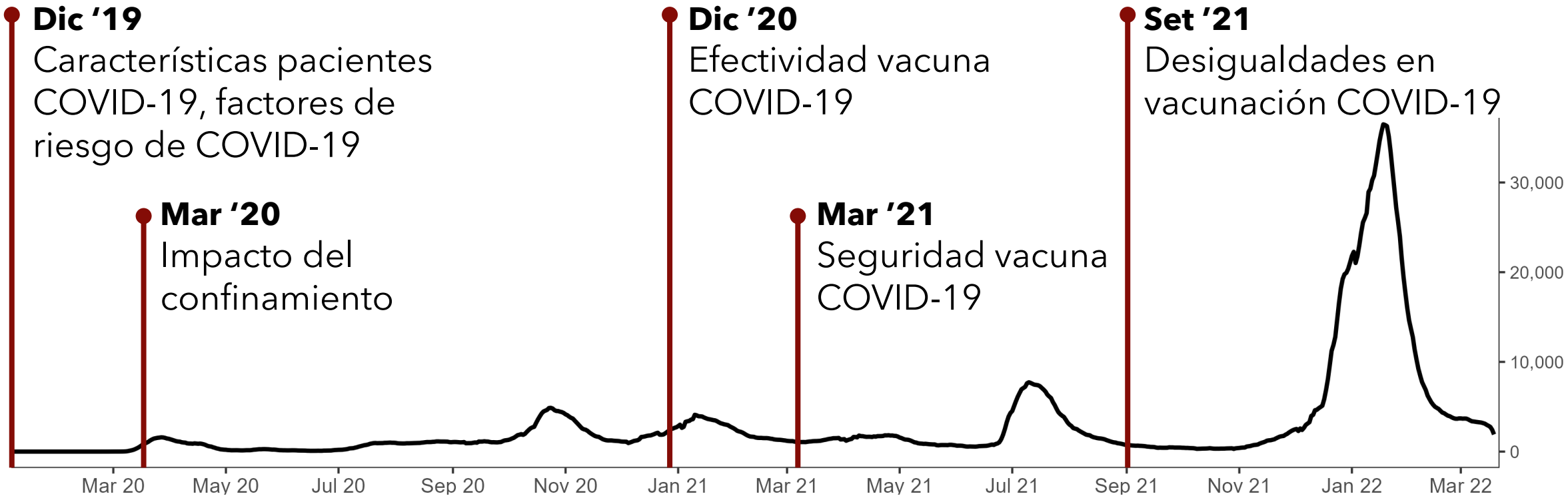
Desigualdades en vacunación COVID-19 en Cataluña

Desigualdades en vacunación en personas de 40-64 años por nivel de privación (10 puntos porcentuales entre Q1 y Q5)

Cobertura vacunal, por nivel de privación y edad.
Q5: mayor privación



Los datos del mundo real aportaron evidencia sobre…



Conclusiones

Durante la pandemia, los datos del mundo real han sido claves para generar evidencia y guiar los esfuerzos en Salud Pública, gracias a sus características:

- Disponibilidad
- Volumen de datos
- Representatividad
- Heterogeneidad

Conclusiones

Durante la pandemia, los datos del mundo real han sido claves para generar evidencia y guiar los esfuerzos en Salud Pública, gracias a sus características:

- Disponibilidad
- Volumen de datos
- Representatividad
- Heterogeneidad

Conclusiones

Los datos del mundo real también tienen limitaciones importantes

Calidad	Incompletos
Riesgo de sesgos	Heterogeneidad